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Methods and apparatus of path switch based service continuity for ue-to- network relay

Abstract

Apparatus and methods are provided for path switch with service continuity for the UE from the indirect relay link to direct cell link or vice versa. In one novel aspect, the UE is configured with measurement report configuration, which includes triggering event for sending the measurement report. The network sends path switch command to the UE to initiate the path switch based on the measurement report. The UE path switch includes intra-gNB and inter-gNB path switch between the indirect relay link and the direct cellular link. The triggering events are a measurement of the target direct Uu link is an offset better than a measurement of the serving sidelink for a indirect link to direct link switch and a measurement of the PC5 link is better than a predefined threshold for direct link to indirect link switch.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is filed under 35 U.S.C. § 111(a) and is based on and hereby claims priority under 35 U.S.C. § 120 and § 365(c) from International Application No. PCT/CN2021/075459, titled “Methods and Apparatus of Path Switch based Service Continuity for UE-to-Network Relay,” with an international filing date of Feb. 5, 2021. Internal Application No. PCT/CN2021/075459, in turn, claims priority under 35 U.S.C. § 120 and § 365(c) from International Application No. PCT/CN2020/074410, titled “Methods and Apparatus of Path Switch based Service Continuity for UE-to-Network Relay,” with an international filing date of Feb. 6, 2020. This application is a continuation of International Application No. PCT/CN2021/075459. International Application No. PCT/CN2021/075459 is pending as of the filing date of this application, and the United States is an elected state in International Application No. PCT/CN2021/075459. The disclosure of each of the foregoing documents is incorporated herein by reference.

TECHNICAL FIELD
(1) The disclosed embodiments relate generally to wireless communication, and, more particularly, to path switch-based service continuity for UE-to-Network relay.
BACKGROUND

(2) 5G radio access technology will be a key component of the modern access network. It will address high traffic growth and increasing demand for high-bandwidth connectivity. Wireless relay in cellular networks provides extended coverage and improved transmission reliability. Long term evolution (LTE) network introduced 3GPP sidelink, the direct communication between two user equipment (UEs) without signal relay through a base station. In 3GPP New Radio (NR), sidelink continues evolving. With new functionalities supported, the sidelink offers low latency, high reliability and high throughout for device to device communications. Using sidelink for wireless relay provides a reliable and efficient way for traffic forwarding. For the early sidelink-based wireless relay services, such as the ProSe UE-to-Network relay, the traffic between the remote UE and the base station is forwarded at the IP layer by the relay UE. The relay operation was specified for LTE aiming at coverage expansion from the perspective of Layer-3 (L3) relay. The Layer-2 (L2) based relay using sidelink can improve the efficiency and flexibility. With the adaption of the sidelink relay, the path switch between direct cellular link and the sidelink relay link, and vice versa are new use cases not addressed by the current network design. The issues of maintain the service continuity during the path switch needs to be addressed.
(3) Improvements and enhancements are required for path switch for service continuity for the UE-to-Network relay.
SUMMARY
(4) Apparatus and methods are provided for path switch with service continuity for the UE from the indirect relay link to direct cell link or vice versa. In one novel aspect, the UE is configured with measurement report configuration, which includes one or more triggering events for sending the measurement report. The network sends path switch command to the UE to initiate the path switch based on the measurement report. The UE path switch includes intra-gNB and inter-gNB path switch between the indirect relay link and the direct cellular link.
(5) In one embodiment, the UE establishes an indirect connection with a source gNB through a sidelink relay path in the NR wireless network, wherein the sidelink relay path has at least one serving relay UE connecting with the UE through a serving sidelink, receives a measurement configuration including a frequency list for measurement from the source gNB, wherein the frequency list including a list of one or more Uu direct links and one or more PC5 sidelinks, sends a measurement report upon detecting a predefined path switching triggering event based on the received measurement configuration, receives a path switching message from the source gNB indicating to switch from the sidelink relay path to a target direct Uu link, and performs a path switching with service continuity by establishing a connection with the target direct Uu link based on the path switching message. In one embodiment, the predefined path switching triggering event is a measurement of the target direct Uu link is an offset better than a measurement of the serving sidelink with the serving relay UE. In another embodiment, the path switching message is a RRC Reconfiguration message from the source gNB.
(6) In one embodiment, the UE with a source Uu link in the NR wireless network, receives from the source gNB, a message for one or more path switching triggering events, sends a response to the source gNB upon detecting at least one path switching triggering event, receives a path switching message from the source gNB indicating to switch from the source Uu link to a sidelink relay path, and performs a path switching with service continuity by establishing a sidelink with a relay UE in the sidelink relay path based on the path switching message. In one embodiment, the message for one or more path switching triggering events is a measurement configuration, and wherein the one or more path switching triggering events comprising a measurement of the PC5 link is an offset better than the source Uu link, and a measurement of the PC5 link is better than a predefined threshold. In another embodiment, the response to the source gNB is a measurement report. In another embodiment, the message for one or more path switching triggering events is a Uu RRC message of a Relay Discovery Command that includes an ID of the relay UE. In another embodiment, the performing the path switch involves receiving, from the relay UE, a PC5 RRC

message of a layer-2 (L2) RLC Channel Establishment Request including a configuration for one or more RLC relay channels over the sidelink.
(7) This summary does not purport to define the invention. The invention is defined by the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The accompanying drawings, where like numerals indicate like components, illustrate embodiments of the invention.
(2) FIG. **1** is a schematic system diagram illustrating exemplary NR wireless network for path switch based service continuity in accordance with embodiments of the current invention.
(3) FIG. **2** illustrates an exemplary NR wireless system with centralized upper layers of the NR radio interface stacks in accordance with embodiments of the current invention.
(4) FIG. **3** illustrates an exemplary Intra-gNB path switch scenario from the indirect relay link to the direct cell link in accordance with embodiments of the current invention.
(5) FIG. **4** illustrates an exemplary Intra-gNB path switch scenario from the direct cell link to the indirect relay link in accordance with embodiments of the current invention.
(6) FIG. **5** illustrates an exemplary Inter-gNB path switch scenario from the indirect relay link to the direct cell link with a different gNB in accordance with embodiments of the current invention.
(7) FIG. **6** illustrates an exemplary Inter-gNB path switch scenario from the direct cell link to the indirect relay link with a different gNB in accordance with embodiments of the current invention.
(8) FIG. **7** illustrates an exemplary flow diagram of an Intra-gNB path switch from the indirect relay link to the direct cell link in accordance with embodiments of the current invention.
(9) FIG. **8** illustrates an exemplary flow diagram of an Intra-gNB path switch from the direct cell link to the indirect relay link in accordance with embodiments of the current invention.
(10) FIG. **9** illustrates an exemplary flow diagram of an Inter-gNB path switch from the indirect relay link to the direct cell link with a different gNB in accordance with embodiments of the current invention.
(11) FIG. **10** illustrates an exemplary of the Inter-gNB path switch from the direct cell link to the indirect relay link with a different gNB in accordance with embodiments of the current invention.
(12) FIG. **11** illustrates an exemplary flow chart for a path switch from an indirect relay link to a direct Uu link with service continuity in accordance with embodiments of the current invention.
(13) FIG. **12** illustrates an exemplary flow chart for a path switch from a direct Uu link to an indirect relay link with service continuity in accordance with embodiments of the current invention.
DETAILED DESCRIPTION
(14) Reference will now be made in detail to some embodiments of the invention, examples of which are illustrated in the accompanying drawings.
(15) FIG. **1** is a schematic system diagram illustrating exemplary NR wireless network for path switch-based service continuity in accordance with embodiments of the current invention. NR wireless system **100** includes one or more fixed base infrastructure units forming a network distributed over a geographical region. The base unit may also be referred to as an access point, an access terminal, a base station, a Node-B, an eNode-B (eNB), a gNB, or by other terminology used in the art. The network can be homogeneous network or heterogeneous network, which can be deployed with the same frequency or different frequency. gNB **101**, gNB **102** and gNB **103** are base stations in the NR network, the serving area of which may or may not overlap with each other. Backhaul connections such as **131**, **132**, and **133**, connect the non-co-located receiving base units, such as gNB **101**, **102** and **103**. These backhaul connections can be either ideal or non-ideal. gNB **101** connects with gNB **102** via Xnr interface **131** and connects with gNB **103** via Xnr interface **132**. gNB **102** connects with gNB **103** via Xnr interface **133**.

(16) NR wireless network **100** also includes multiple communication devices or mobile stations, such user equipments (UEs) such as UEs **111**, **112**, **113**, **117**, **118**, **121**, **122**, **123**, and **128**. Communication devices or mobile stations in NR wireless network **100** may also be referred to as a mobile station, a mobile terminal, a mobile phone, smart phone, wearable, an IoT device, a table let, a laptop, or other terminology used in the art. It may also refers to devices with wireless connectivity in a vehicle, such as mobile devices **118**, **117** and **128**. The exemplary mobile devices in wireless network **100** have sidelink capabilities. The mobile devices can establish one or more connections with one or more base stations, such as gNB **101**, **102**, and **103**. The mobile device, such as mobile device **117**, may also be out of connection with the base stations with its access links but can transmit and receive data packets with another one or more other mobile stations or with one or more base stations through L2-based sidelink relay.
(17) Data packets are forwarded by one or more relay UEs based on information in L2 header. A remote UE **111** and gNB **103** forms an end-to-end path **181** through a L2-based sidelink relay with a relay UE **121**. End-to-end path **181** includes an access link **135** between gNB **103** and relay UE **121** and a sidelink **171** between remote UE **111** and relay UE **121**. In one embodiment, the remote UE **111** also has a direct access link **138** with gNB **103**. In other embodiments, the L2-based sidelink relay is a multi-hop relay with multiple relay UEs. A remote UE **112** and gNB **102** forms an end-to-end path **182** through a L2-based sidelink relay with a relay UE **122** and another relay UE **123**. End-to-end path **182** includes an access link **136** between gNB **102** and relay UE **122**, sidelink **172** between relay UE **122** and relay UE **123**, and sidelink **173** between remote UE **112** and relay UE **123**. In yet another embodiment, a relay mobile device is configured with multiple remote mobile devices or multiple end node mobile devices. A relay UE **128**, with an access link **137** to gNB **101** is configured with two remote UEs **117** and **118** through sidelink **175** and **176**, respectively. In one embodiment, the UE-to-Network L2-based sidelink relay includes one or more remote UEs being the out-of-coverage UEs, such as remote UE **117**. Different links are established for the illustrated relay paths. An access link is a link between a base station, such as gNB and a mobile device, such as a UE. The UE can be a remote UE or a relay UE. The access link includes both the uplink (UL) and the downlink (DL) between the base station and the mobile device. The interface for the access link is an NR Uu interface. In one embodiment, the remote UE also establishes access link with a base station, such as remote UE **111** establishes the access link **138** with gNB **103**. A sidelink is a link between two mobile devices and uses PC5 interface. The sidelink can be a link between a remote UE/end-node UE and a relay UE or a link between two relay mobile devices/UEs for the multi-hop relay. The end-to-end link for a relay path can be a link between two end-node mobile devices for a UE-to-UE relay or a base station to mobile device for a UE-to-Network relay. An Xn link is the backhaul link between two base stations, such gNBs using the Xn interface. In one embodiment, candidate relay UE information is transmitted to the base station via the Xn link.
(18) In one novel aspect, for a UE-to-Network connection, path switch between the direct Uu/cellular connection and the indirect link through sidelink relay is performed with service continuity. For example, UE **121** is connected with gNB **103** via sidelink relay through path **181** which includes the sidelink **171** and Uu direct link **138**. A path switch is triggered upon one or more predefined triggering events. UE **121** performs the path switch to the Uu direct link **135** with service continuity. In a different scenario, when UE **121** is connected with gNB **103** with direct link **135**, it performs path switch to the relay path **181** with sidelink relay through UE **111**. The path switch is performed with service continuity.
(19) FIG. **1** further illustrates simplified block diagrams of a base station and a mobile device/UE for adaptation handling for L2-based sidelink relay. gNB **103** has an antenna **156**, which transmits and receives radio signals. An RF transceiver circuit **153**, coupled with the antenna, receives RF signals from antenna **156**, converts them to baseband signals, and sends them to processor **152**. RF transceiver **153** also converts received baseband signals from processor **152**, converts them to RF

signals, and sends out to antenna **156**. Processor **152** processes the received baseband signals and invokes different functional modules to perform features in gNB **103**. Memory **151** stores program instructions and data **154** to control the operations of gNB **103**. gNB **103** also includes a set of control modules **155** that carry out functional tasks to communicate with mobile stations.

(20) FIG. **1** also includes simplified block diagrams of a UE, such as relay UE **121** or remote UE **111**. The UE has an antenna **165**, which transmits and receives radio signals. An RF transceiver circuit **163**, coupled with the antenna, receives RF signals from antenna **165**, converts them to baseband signals, and sends them to processor **162**. In one embodiment, the RF transceiver may comprise two RF modules (not shown). A first RF module is used for HF transmitting and receiving, and the other RF module is used for different frequency bands transmitting and receiving which is different from the HF transceiver. RF transceiver **163** also converts received baseband signals from processor **162**, converts them to RF signals, and sends out to antenna **165**. Processor **162** processes the received baseband signals and invokes different functional modules to perform features in the UE. Memory **161** stores program instructions and data **164** to control the operations of the UE. Antenna **165** sends uplink transmission and receives downlink transmissions to/from antenna **156** of gNB **103**.

(21) The UE also includes a set of control modules that carry out functional tasks. These control modules can be implemented by circuits, software, firmware, or a combination of them. An original path handler **191** establishes an indirect connection with a source gNB through a sidelink relay path in the NR wireless network, wherein the sidelink relay path has at least one serving relay UE connecting with the UE through a serving sidelink. A configuration receiver **192** receives a measurement configuration including a frequency list for measurement from the source gNB, wherein the frequency list including a list of one or more Uu direct links and one or more PC5 sidelinks. A response handler **193** sends a measurement report upon detecting a predefined path switching triggering event based on the received measurement configuration. A command receiver **194** receives a path switching message from the source gNB indicating to switch from the sidelink relay path to a target direct Uu link. A path switch handler **195** performs a path switching with service continuity by establishing a connection with the target direct Uu link based on the path switching message.

(22) FIG. **2** illustrates an exemplary NR wireless system with centralized upper layers of the NR radio interface stacks in accordance with embodiments of the current invention. Different protocol split options between central unit (CU) and distributed unit (DU) of gNB nodes may be possible. The functional split between the CU and DU of gNB nodes may depend on the transport layer. Low performance transport between the CU and DU of gNB nodes can enable the higher protocol layers of the NR radio stacks to be supported in the CU, since the higher protocol layers have lower performance requirements on the transport layer in terms of bandwidth, delay, synchronization and jitter. In one embodiment, SDAP and PDCP layer are located in the CU, while RLC, MAC and PHY layers are located in the DU. A core unit **201** is connected with one central unit **211** with gNB upper layer **252**. In one embodiment **250**, gNB upper layer **252** includes the PDCP layer and optionally the SDAP layer. Central unit **211** connects with distributed units **221**, **222**, and **221**. Distributed units **221**, **222**, and **223** each corresponds to a cell **231**, **232**, and **233**, respectively. The DUs, such as **221**, **222** and **223** includes gNB lower layers **251**. In one embodiment, gNB lower layers **251** include the PHY, MAC and the RLC layers. In another embodiment **260**, each gNB has the protocol stacks **261** including SDAP, PDCP, RLC, MAC and PHY layers.

(23) The communication based on UE-to-Network Relay requires service continuity during communication path switch between a direct Uu path and an indirect Uu path via a UE-to-Network relay. There are different scenarios for the UE-to-Network relay path switch with sidelink relay.

(24) FIG. **3** illustrates an exemplary Intra-gNB path switch scenario from the indirect relay link to the direct cell link in accordance with embodiments of the current invention. A UE **303** is connected to a gNB **301** via an indirect link through a relay UE **302**. The indirect link relay path

includes the sidelink **312** between UE **303** and relay UE **302**, and a Uu link **311** between relay UE **302** and gNB **301**. In one embodiment, UE **303** receives measurement report and performs measurement on the relay link and direct Uu link. UE **303** is configured with triggering event to send the measurement report. The network upon receiving the measurement report from UE **303**, determines to initiate the path switch. UE **303** receives path switch command. At step **321**, UE switches path to Uu direct link **313** with gNB **301**.

(25) FIG. **4** illustrates an exemplary intra-gNB path switch scenario from the direct cell link to the indirect relay link in accordance with embodiments of the current invention. A UE **403** is connected to a gNB **401** via a direct Uu link **413**. In one embodiment, the UE **403** receives measurement report configuration from the network including frequency list of both the Uu link with gNB and sidelink with relay UE. UE **403** performs PC5-S discovery with relay UE **402**. Upon detecting predefined triggering event, UE **403** sends measurement report to the network. The network determines to perform a path switch. UE **403** receives path switch command from the network. At step **421**, UE **403** switches path to indirect link with gNB **401**. The indirect link includes a sidelink **412** between UE **403** and relay UE **402** and a Uu direct link **411** between relay UE **402** and gNB **401**. In another embodiment, UE **403** receives a relay discovery command from gNB **401** and performs the PC5-S discovery with relay UE **402**. UE **403** sends Sidelink UE Information (SUI) message to gNB **401**. The network determines to perform a path switch. UE **403** receives path switch command from the network. At step **421**, UE **403** switches path to indirect link with gNB **401**. The indirect link includes a sidelink **412** between UE **403** and relay UE **402** and a Uu direct link **411** between relay UE **402** and gNB **401**.

(26) FIG. **5** illustrates an exemplary inter-gNB path switch scenario from the indirect relay link to the direct cell link with a different gNB in accordance with embodiments of the current invention. A UE **503** is connected to a gNB **501** via an indirect link through a relay UE **502**. The indirect link relay path includes the sidelink **512** between UE **503** and relay UE **502**, and a Uu link **511** between relay UE **502** and gNB **501**. In one embodiment, UE **503** receives measurement report and performs measurement on the relay link and direct Uu link, including Uu link with a different gNB **504**. UE **503** is configured with triggering event to send the measurement report. The network upon receiving the measurement report from UE **503**, determines to initiate the path switch. UE **503** receives path switch command. At step **521**, UE switches path to Uu direct link **513** with gNB **504**.

(27) FIG. **6** illustrates an exemplary inter-gNB path switch scenario from the direct cell link to the indirect relay link with a different gNB in accordance with embodiments of the current invention. A UE **603** is connected to a gNB **604** via a direct Uu link **613**. In one embodiment, the UE **603** receives measurement report configuration from the network including frequency list of both the Uu link with gNB and sidelink with relay UE. UE **603** performs PC5-S discovery with relay UE **602**. Upon detecting predefined triggering event, UE **603** sends measurement report to the network. The network determines to perform a path switch. UE **603** receives path switch command from the network. At step **621**, UE **603** switches path to an indirect link via relay UE **602** to different gNB **601**. The indirect link includes a sidelink **612** between UE **603** and relay UE **602** and a Uu direct link **611** between relay UE **602** and gNB **601**. In another embodiment, UE **603** receives a relay discovery command from gNB **604** and performs the PC5-S discovery with relay UE **602**. UE **603** sends Sidelink UE Information (SUI) message to gNB **604**. The network determines to perform a path switch. UE **603** receives path switch command from the network. At step **621**, UE **603** switches path to an indirect link via relay UE **602** to different gNB **601**. The indirect link includes a sidelink **612** between UE **603** and relay UE **602** and a Uu direct link **611** between relay UE **602** and gNB **601**.

(28) In one novel aspect, the sidelink relay for the path switch is layer-2 UE-to-network relay. When the relay UE and the remote UE established a PC5 unicast link, the relay UE and the remote UE are associated to perform traffic relaying. A remote UE-relay UE association, or relay UE-remote UE association is, thereby, formed. A pair of UE identities (e.g. UE L2 ID, index of UE ID

or truncated version of UE ID) indicate this type of association.

(29) FIG. **7** illustrates an exemplary flow diagram of an intra-gNB path switch from the indirect relay link to the direct cell link in accordance with embodiments of the current invention. At step **711**, there is an ongoing indirect traffic between a remote UE **701** and gNB **703** through a relay UE **702** in the NR wireless network with a core network (CN) entity **704**. At step **721**, gNB **703** sends the measurement configuration, via the relay link through relay UE **702**, to remote UE **701** including the frequency list to be measured for the cellular link and/or relay link. At step **722**, remote UE **701** sends Uu RRC message (e.g. Measurement Report) to the gNB **703** via the UE-to-Network Relay. The sending of the measurement report is triggered by one or more triggering events. In one embodiment, the triggering event is configured by the measurement report configuration at step **721**. In another embodiment, a new A3-link event is defined as “cellular quality (Uu) offset better than serving relay UE (PC5)”. UE **701** determines whether one or more triggering events are met based on its measurement results. At step **731**, based on received RRC message, the measurement report, gNB **703** decides to handover the remote UE **701** from the UE-to-Network relay to the serving gNB **703**.

(30) At step **751**, gNB **703** sends RRCReconfiguration message via the UE-to-Network relay to the remote UE **701** to configure the PHY, MAC, RLC, PDCP, and SDAP layers of the link between remote UE **701** and gNB **703**. In another embodiment, the RRCReconfiguration message only configures the PHY, MAC and RLC layers of the link between remote UE **701** and gNB **703**. From L2 relaying perspective, the PDCP and SDAP layer for remote UE terminate at remote UE **701** and gNB **703**, there is no change on the PDCP/SDAP configuration after path switch. At step **752**, remote UE **701** sends Reconfiguration Complete message gNB **703**. At step **761**, Uu interface (i.e. the direct path) can carry the traffic between remote UE **701** and gNB **703**.

(31) At step **771**, gNB **703** and CN **704**, such as an access and mobility management function (AMF), exchanges signaling to update the UE context. It is assumed both gNB and AMF (relay UE's AMF) stores remote UE-relay UE association within the relay UE context. The AMF is updated on the change of the association status (i.e. the release of the remote UE-relay UE association because of the PC5.fwdarw.Uu Switch). gNB **703** makes synchronized changes on the remote UE-relay UE association together with AMF. The status of remote UE-relay UE association is sent to remote UE's AMF in order to track the UE from mobility point of view.

(32) At step **781**, gNB **703** sends RRCReconfiguration message to the relay UE **702** to configure the PHY, MAC, RLC, PDCP, SDAP layers or any combination among them for the link between relay UE and base station. In one embodiment, the RRCReconfiguration message only configures the release list of the DRBs, for example only DRB-ToReleaseList is included within the IE RadioBearerConfig or RelayRadioBearerConfig. In another embodiment, the RRCReconfiguration message only indicates the index of remote UE-relay UE association, which implicitly means release all the DRBs established for relaying for this remote UE-relay UE association. For example, only a remote UE-relay UE association index is included in the DRB-ToReleaseList within the IE RadioBearerConfig or RelayRadioBearerConfig included in RRCReconfiguration message. At step **782**, relay UE **702** sends RRCReconfigurationComplete message to the base station **703** when the reconfiguration completes. At step **791**, the PC5 unicast link between remote UE **701** and the relay UE **702** is released via PC5-S signaling. The PC5 RRC connection between remote UE **701** and the relay UE **702** is released automatically.

(33) FIG. **8** illustrates an exemplary flow diagram of an Intra-gNB path switch from the direct cell link to the indirect relay link in accordance with embodiments of the current invention. At step **811**, there is an ongoing direct traffic between a remote UE **801** and gNB **803** in the NR wireless network with a core network (CN) entity **704**. At step **812**, relay UE **802** is connected with remote UE's serving gNB **803**. At step **813**, relay UE **802** sends a relay indication to its serving gNB via an RRC message, e.g. SidelinkUEInformation. It indicates to gNB **803** that relay UE **802** can provide UE-to-Network relay in a certain frequency list. For example, during V2X platooning operation,

the head node takes the responsibility to forward the member node's Uu traffic. In a scenario where relay UE allocated the remote UE's L2 ID before, the relay UE can include L2 UE ID of one or more remote UEs into the RRC message, e.g. SidelinkUEInformation, to inform the base station the one or more remote UEs the relay UE can establish relay link to.

(34) In one embodiment, at step **821**, gNB **803** sends the measurement configuration to remote UE **801** including the frequency list to be measured for cellular link and/or relay link. In another embodiment, at step **822**, instead of the measurement configuration, the gNB **803** sends Uu RRC message, e.g. Relay Discovery Command to remote UE **801**. This can be triggered by an indication previously sent from remote UE to base station showing the interest to operate as remote UE via UE-to-Network relay in a certain frequency list. This can be also triggered by a relay indication sent from a relay UE at step **813** to the serving gNB **803**, The RRC message of Relay Discovery Command includes relay UE identity information (e.g. L2 ID), which helps remote UE **801** to find the right relay UE for discovery. In another embodiment, the RRC message of Relay Discovery Command carries the content of measurement configuration message.

(35) At step **823**, remote UE **801** discovers relay UE **802** and selects relay UE **802** via PC5-S signaling. The triggered PC5-S relay discovery is based on the measurement event as configured in measurement configuration at step **821**, or based on the criteria configured via system information, or (pre-)configuration for relay selection is met. Steps **821** and **822** configure the NR existing measurement event. In another embodiment, a new measurement event or criteria can be configured. This new event or criteria can be defined as “Relay UE quality (PC5) offset better than serving cell (Uu)”, “Relay UE quality (PC5) better than an absolute threshold”, or “Relay UE quality (PC5) offset better than an absolute threshold”. Remote UE **801** evaluates the event or criteria for each measurement result before selecting the relay **802**. If a relay UE L2 ID is provided at step **822** to remote UE **801**, remote UE puts this ID into the PC5-S discovery message, e.g. a broadcasted Direct Communication Request, in order to establish the PC5 unicast link with the right relay UE. The expected effect is that only the right relay UE responds Direct Communication Response to remote UE **801**. As an additional option, Remote UE's own L2 ID can be put into the PC5-S message for discovery purpose e.g. a broadcasted Direct Communication Request, in order to allow the Relay UE to determine his response based on upper layer or application layer restriction to the received Direct Communication Request. At step **824**, remote UE sends Uu RRC message, e.g. Measurement Report, to the base station. In another embodiment, at step **825**, remote UE **801** sends a Sidelink UE information message to the gNB **803**. The Relay UE Identity information, e.g. Relay UE ID, is reported to the gNB **803**. At step **831**, based on the received RRC message, gNB **803** decides path switch for remote UE **801** to handover from the serving gNB **803** to the UE-to-Network Relay.

(36) At step **851**, gNB **803** sends RRCReconfiguration message to remote UE **801**. gNB **803** instructs remote UE **801** to switch from the direct Uu PHY radio resource and to release the corresponding RLC entity and MAC entity for Uu based Radio Bearer(s). The SDAP and PDCP configuration of the Uu based Radio Bearer(s) is kept unchanged. gNB **803** instructs remote UE **801** to establish the unicast link with relay **802**. At step **852**, remote UE **801** establishes PC5-S link with relay UE **802**. In one embodiment, the security is activated after the establishment of the unicast link between remote UE **801** and relay UE **802**. At step **853**, remote UE **801** sends RRCReconfigurationComplete message to the gNB **802**.

(37) At step **881**, gNB **803** sends RRCReconfiguration message to the relay UE **802** to configure the PHY, MAC, RLC, PDCP, SDAP layers or any combination among them for the relaying bearer(s) between relay UE **802** and gNB **803**. The existing RadioBearerConfig structure or a new defined RelayRadioBearerConfig structure is used to hold all of the Radio Bearer (i.e. RB) configuration for relaying. In one embodiment, relay UE **802** serves multiple remote UEs. Both relay UE **803** and its serving base station, gNB **803**, maintain multiple relay UE-remote UE associations. The serving gNB can allocate one index for each relay UE-remote UE association for

the relay UE. If the existing RadioBearerConfig structure is used to express the relaying RB configuration, additional fields (e.g. relaying RB indicator, index of each relay UE-remote UE association, remote UE ID) is needed to distinguish relaying RB from normal RB, and to distinguish different RBs for each relay UE-remote UE association. If a new defined RelayRadioBearerConfig structure is used to express the relaying RB configuration, additional fields (e.g. index of each relay UE-remote UE association, Remote UE ID) is needed to distinguish different RBs for each Relay UE-Remote UE association. In one embodiment, the SDAP-Config for relaying RB is omitted during RRC Reconfiguration, the establishment of relaying RB. When performing relay association, gNB maps the end-to-end RB to indirect RB (i.e. relaying RB) between relay UE and gNB and the flow concept is not applied any more after bearer mapping. Specifically, when multiple end-to-end RBs are mapped into one indirect RB, multiple SDAP-config in one indirect RB does not apply. The relaying RB configuration includes PDCP-config, while the cnAssociation IE, which is currently mandatory for DRB setup, is omitted. In one embodiment, the existing RadioBearerConfig structure or a new defined RelayRadioBearerConfig structure used to configure relaying RB(s) can include one or any combination of the following fields: Relaying RB indicator drb-Identity pdcp-Config Index of Relay UE-Remote UE association Relay UE ID Remote UE ID Mapped end-to-end drb-Identity L2 RLC channel ID of Sidelink for relaying Logical channel ID of L2 RLC relay channel

(38) At step **882**, upon receiving the RRCReconfiguration message from gNB **803**, relay UE **802** initiates L2 RLC channel establishment procedure for relaying between relay UE **802** and remote UE **801** by sending L2 RCL Channel Establishment Request message. This RRC message includes one or a list of RLC Relay channel configuration. Each relay channel configuration includes the associated end-to-end Uu radio bearer index(es) (e.g., ServedRadioBearer), RLC layer index (e.g. sl-RLC-BearerConfigIndex), RLC configuration (e.g. sl-RLC-Config), MAC-logical channel configuration (e.g. sl-MAC-LogicalChannelConfig), sidelink logical channel ID, relay channel ID, the QoS profile of the QoS flow(s) within each associated end-to-end Uu radio bearer, or any combination among them. The RLC configuration includes RLC mode, SN length, etc. In another embodiment, the relay UE-remote UE association is also included in the message to indicate that the relay Channel is established for a relay UE-remote UE association. In one embodiment, the list of end-to-end radio bearer IDs is included for each relay channel within each relay channel configuration. It informs the remote UE the RLC channel that carries a particular end-to-end radio bearer. At step **882**, remote UE **801** sends a PC5 RRC message, e.g. L2 RLC Channel Establishment Complete, to relay UE **802** to acknowledge the establishment of L2 RLC channel for relaying. In another embodiment, step **882** is triggered by step **851**. The L2 RLC Channel Establishment Request is sent from remote UE **801** to relay UE **802** and relay UE **802** acknowledges L2 RLC Channel Establishment Complete to remote UE **801** when L2 RLC channel establishment completes. The RRC reconfiguration message at **851** carries the configuration of L2 RLC channel for relaying. At step **884**, relay UE **802** sends RRCReconfigurationComplete message to gNB **803**. At step **891**, PC5 can carry the traffic between remote UE and network.

(39) FIG. **9** illustrates an exemplary flow diagram of an inter-gNB path switch from the indirect relay link to the direct cell link with a different gNB in accordance with embodiments of the current invention. At step **911**, there is an ongoing indirect traffic between a remote UE **901** and gNB **903** through a relay UE **902** in the NR wireless network with a core network (CN) entity **905**. At step **921**, gNB **903** sends the measurement configuration, via the relay link through relay UE **902**, to remote UE **901** including the frequency list to be measured for the cellular link and/or relay link. At step **922**, remote UE **901** sends Uu RRC message (e.g. Measurement Report) to the gNB **903** via the UE-to-Network Relay. The sending of the measurement report is triggered by one or more triggering events. In one embodiment, the triggering event is configured by the measurement report configuration at step **921**. In another embodiment, a new A3-link event is defined as “cellular quality (Uu), including Uu link with a different gNB, offset better than serving Relay UE

(PC5)”. UE **901** determines whether one or more triggering events are met based on its measurement results. At step **931**, based on received RRC message, the measurement report, gNB **903** decides to handover the remote UE **901** from the UE-to-Network relay to the serving gNB **903**. At step **951**, after the serving gNB **903** of remote UE **901** decides to switch the remote UE from relay link to cellular link, the source gNB **903** initiates HO request to the target gNB **904** for remote UE **901** via Xn interface. In another embodiment, the source gNB initiates HO request to a source CN entity **905**, such as a source AMF, via N2 interface. Source AMF sends a HO required to the target AMF. The target AMF sends HO request to the target gNB **904** for remote UE **901**. At step **951**, the target gNB **904** responses the HO acknowledgement message via Xn interface. Alternatively, target gNB **904** responses the HO acknowledgement message to the target AMF via N2 interface. The target AMF responses HO acknowledgement to source AMF. The source AMF sends HO acknowledgement to the source gNB **903** for remote UE **901**.

(40) At step **961**, the source gNB **903** delivers the Handover Command information to remote UE **901** via RRC reconfiguration message, which includes mobilitycontrolinfo. At step **962**, the source gNB **903** sends SN Status Transfer to the target gNB **904**. At step **963**, the source gNB **903** forwards the data to the target gNB **904**. At step **964**, remote UE **901** sends RRC Reconfiguration Complete message to the target gNB **904**. At step **965**, the target gNB **904** sends a PATH SWITCH REQUEST message to AMF (in CN **905**) to inform that the UE has changed cell. The path switch request may also update the remote UE's context, e.g. the status of remote UE-relay UE association in the AMF. At step **966**, the AMF confirms the path switch message with the Path Switch Request Acknowledge message to target gNB **904**. At step **967**, the target gNB **904** notifies the source gNB **903** to release the UE context of remote UE **901**. At step **971**, the direct Uu interface between remote UE **901** and target gNB **904** carries the direct traffic between remote UE and network. At step **981**, the source gNB of relay UE **902** (assuming the same source gNB for both remote UE and relay UE in this scenario) sends RRCReconfiguration message to the UE-to-Network relay UE **902**. At step **982**, relay UE **902** sends RRCReconfigurationComplete message to source gNB **903**. At step **991**, the PC5 unicast link between remote UE **901** and the relay UE **902** is released via PC5-S signaling. The PC5 RRC connection between remote UE **901** and the relay UE **902** is released automatically.

(41) FIG. **10** illustrates an exemplary of the inter-gNB path switch from the direct cell link to the indirect relay link with a different gNB in accordance with embodiments of the current invention. At step **1011**, there is an ongoing direct traffic between a remote UE **1001** and gNB **1003** in the NR wireless network with a core network (CN) entity **1005**. At step **1012**, relay UE **1002** is connected with a target gNB **1004**. At step **1013**, relay UE **1002** sends a Relay indication to its serving gNB, target gNB **1004**, via a RRC message e.g. SidelinkUEInformation. It indicates to gNB **1004** that relay UE **1002** can provide UE-to-Network relay in a certain frequency list.

(42) In one embodiment, at step **1021**, gNB **1003** sends the measurement configuration to remote UE **1001** including the frequency list to be measured for cellular link and/or relay link. In another embodiment, at step **1022**, instead of the measurement configuration, the gNB **1003** sends Uu RRC message, e.g. Relay Discovery Command to remote UE **1001**. This can be triggered by an indication previously sent from remote UE to base station showing the interest to operate as remote UE via UE-to-Network relay in a certain frequency list. In another embodiment, the RRC message of Relay Discovery Command carries the content of measurement configuration message.

(43) At step **1023**, remote UE **1001** performs relay discovery and selection. During the relay discovery stage, remote UE **1001** gets the serving cell ID of target gNB **1004** of relay UE **1002**. The discovery between remote UE **1001** and relay UE **1002** is managed by PC5-S signaling. If no explicit relay discovery procedure is supported before sidelink communication, relay UE **1002** puts its serving cell ID of gNB **1004** into the PC5-S Direct Communication Response when accepting the PC5-S Direct Communication Request from remote UE **1001**. If explicit relay discovery procedure proceeds before sidelink communication, relay UE puts its serving cell ID of gNB **1004**

into the PC5-S Discovery Announcement (in discovery mode A) or Discovery Response (in discovery mode B). In one embodiment, remote UE **1001** and relay UE **1002** can put corresponding serving cell ID into the PC5-S signaling exchanged for the purpose of relay discovery. In one embodiment, relay UE **1002** reports the associated remote UE (including both remote UE L2 ID and serving cell ID of remote UE) to its serving gNB **1004** to allow the base station to prepare remote UE access, and/or coordinate with source gNB of remote UE.

(44) At step **1024**, remote UE **1001** sends Uu RRC message, e.g. Measurement Report, to source gNB **1003**. In another embodiment, at step **1025**, remote UE **1001** sends a Sidelink UE information message to the gNB **1003**. The relay UE information and/or relay UE serving cell identity are reported to the target gNB **1004**. The relay UE information includes one or more elements of relay UE L2 ID, C-RNTI, ShortMAC-I, or any combination among them. Relay UE serving cell identity can include CGI and/or PCI. At step **1031**, serving gNB **1003** decides to switch remote UE **1001** from cellular link to relay link. At step **1051**, source gNB **1003** initiates HO procedure. The relay UE-remote UE association is included in the HO Request message. At step **1052**, source gNB **1003** receives the HO acknowledgement, which includes the relay UE-remote UE association. In one embodiment, multiple relay UE-remote UE associations are included in the HO Request and HO Ack messages.

(45) At step **1061**, source gNB **1003** delivers the handover command information to remote UE **1001** via RRC reconfiguration message, which includes mobilitycontrolinfo. gNB **1003** also instructs remote UE **1001** to establish the unicast link with relay UE **1002**. At step **1062**, the source gNB **1003** sends SN Status Transfer to the target gNB **1004**. At step **1063**, source gNB **1003** forwards the data to the target gNB **1004**. At step **1064**, the PC5-S unicast link is established between remote UE **1001** and relay UE **1002**. At step **1065**, remote UE **1001** sends RRCReconfigurationComplete message to the target gNB **1004**. At step **1071** target gNB **1004** sends RRCReconfiguration message to relay UE **1002**. At step **1072**, relay UE **1002** sends a PC5 RRC message, e.g. L2 RLC channel establishment request, to remote UE **1001**. At step **1073**, remote UE **1001** sends a PC5 RRC message e.g. L2 RLC channel establishment complete to relay UE **1002** to acknowledge the establishment of L2 RLC channel for relaying. In another embodiment, step **1072** is triggered by the Handover Command at step **1061**. L2 RLC Channel Establishment Request is sent from remote UE **1001** to relay UE **1002**. Relay UE **1002** acknowledges with L2 RLC Channel Establishment Complete to remote UE **1001**. The Handover Command at step **1061**, in this embodiment, carries the configuration of L2 RLC channel for relaying. At step **1074**, relay UE **1002** sends RRCReconfigurationComplete message to the target gNB **1004**, At step **1081**, PC5 can carry the traffic between remote UE **1001** and Network **1005**. At step **1082**, target gNB **1004** sends a PATH SWITCH REQUEST message to AMF (in CN **1005**) to inform that the UE has changed cell. At step **1083**, the AMF confirms the path switch message with the Path Switch Request Acknowledge message to target gNB **1004**. At step **1091**, target gNB **1004** notifies the source gNB **1003** to release the UE context of remote UE **1001**.

(46) FIG. **11** illustrates an exemplary flow chart for a path switch from an indirect relay link to a direct Uu link with service continuity in accordance with embodiments of the current invention. At step **1101**, the UE establishes an indirect connection with a source gNB through a sidelink relay path in a new radio (NR) wireless network, wherein the sidelink relay path has at least one serving relay UE connecting with the UE through a serving sidelink. At step **1102**, the UE receives a measurement configuration including a frequency list for measurement from the source gNB, wherein the frequency list including a list of one or more Uu direct links and one or more PC5 sidelinks. At step **1103**, the UE sends a measurement report upon detecting a predefined path switching triggering event based on the received measurement configuration. At step **1104**, the UE receives a path switching message from the source gNB indicating to switch from the sidelink relay path to a target direct Uu link. At step **1105**, the UE performs a path switching with service continuity by establishing a connection with the target direct Uu link based on the path switching

message.
(47) FIG. **12** illustrates an exemplary flow chart for a path switch from a direct Uu link to an indirect relay link with service continuity in accordance with embodiments of the current invention. At step **1201**, the UE establishes a direct connection with a source gNB through a source Uu link in a new radio (NR) wireless network. At step **1202**, the UE receives a message for one or more path switching triggering events from the source gNB. At step **1203**, the UE sends a response to the source gNB upon detecting at least one path switching triggering event. At step **1204**, the UE receives a path switching message from the source gNB indicating to switch from the source Uu link to a sidelink relay path. At step **1205**, the UE performs a path switching with service continuity by establishing a sidelink with a relay UE in the sidelink relay path based on the path switching message.
(48) Although the present invention has been described in connection with certain specific embodiments for instructional purposes, the present invention is not limited thereto. Accordingly, various modifications, adaptations, and combinations of various features of the described embodiments can be practiced without departing from the scope of the invention as set forth in the claims.

Claims

1. A method comprising: establishing, by a user equipment (UE), an indirect connection with a source gNB through a sidelink relay path in a new radio (NR) wireless network, wherein the sidelink relay path has at least one serving relay UE connecting with the UE through a serving sidelink; receiving a measurement configuration including a frequency list for measurement from the source gNB, wherein the frequency list including a list of one or more Uu direct links and one or more PC5 sidelinks; sending a measurement report upon detecting a predefined path switching triggering event based on the received measurement configuration; receiving a path switching message from the source gNB indicating to switch from the sidelink relay path to a target direct Uu link; and performing a path switching with service continuity by establishing a connection with the target direct Uu link based on the path switching message.
2. The method of claim 1, wherein the predefined path switching triggering event is a measurement of the target direct Uu link is an offset better than a measurement of the serving sidelink with the serving relay UE.
3. The method of claim 1, wherein the path switching message is a RRC Reconfiguration message from the source gNB.
4. The method of claim 3, wherein the RRC Reconfiguration message includes one or more elements comprising: a list of data radio bearers (DRBs) to be released, a RadioBearerConfig information element (IE) with an index of relay UE-remote UE association, a new IE with an index of relay UE-remote UE association to indicate one or more relay radio bearers to be released.
5. The method of claim 1, wherein the direct target Uu link is with a new gNB, and wherein the path switching message is a RRC Reconfiguration message from the source gNB embedded a Handover Command.
6. A method comprising: establishing, by a user equipment (UE), a direct connection with a source gNB through a source Uu link in a new radio (NR) wireless network; receiving, from the source gNB, a message for one or more path switching triggering events; sending a response to the source gNB upon detecting at least one path switching triggering event; receiving a path switching message from the source gNB indicating to switch from the source Uu link to a sidelink relay path; and performing a path switching with service continuity by establishing a sidelink with a relay UE in the sidelink relay path based on the path switching message.
7. The method of claim 6, wherein the message for one or more path switching triggering events is a measurement configuration, and wherein the one or more path switching triggering events

comprising a measurement of the PC5 link is an offset better than the source Uu link, and a measurement of the PC5 link is better than a predefined threshold.
8. The method of claim 7, wherein the response to the source gNB is a measurement report.
9. The method of claim 7, the message for one or more path switching triggering events is a Uu RRC message of a Relay Discovery Command that includes an ID of the relay UE.
10. The method of claim 9, wherein the response to the source gNB is a Sidelink UE Information (*SUI*) comprising one or more elements comprising the relay UE information and an ID of a serving cell for the relay UE.
11. The method of claim 6, wherein path switching message from the source gNB is a RRC Reconfiguration message that configures relay radio bearer for the sidelink relay path.
12. The method of claim 11, wherein the RRC Reconfiguration message includes one or more elements comprising a drb-Identity, a pdcp-Config, an index of Relay UE-Remote UE association, the UE ID of the Relay UE, a UE ID of the UE, a mapped end-to-end drb-Identity, an L2 RLC channel ID of the relay sidelink, and a Logical channel ID of a L2 RLC relay channel, and wherein the one or more elements are included in one structure selected from an existing RadioBearerConfig, and a new structure of RelayRadioBearerConfig.
13. The method of claim 11, wherein the relay UE is connected with a target gNB, and wherein the RRC Reconfiguration message embedded a Handover Command.
14. The method of claim 6, wherein the performing the path switch involving: receiving, from the relay UE, a PC5 RRC message of a layer-2 (L2) RLC Channel Establishment Request including a configuration for one or more RLC relay channels over the sidelink.
15. The method of claim 14, wherein the configuration for one or more RLC relay channels includes one or more elements comprising: one or more associated end-to-end Uu Radio Bearer indexes, an RLC layer index, an RLC configuration, a MAC-logical channel configuration, a Sidelink logical channel ID, a Relay channel ID, and a QoS profile of one or more QoS flows within each associated end-to-end Uu radio bearer.
16. A user equipment (UE), comprising: a transceiver that transmits and receives radio frequency (RF) signal in a new radio (NR) wireless network; an original path handler that establishes an indirect connection with a source gNB through a sidelink relay path in the NR wireless network, wherein the sidelink relay path has at least one serving relay UE connecting with the UE through a serving sidelink; a configuration receiver that receives a measurement configuration including a frequency list for measurement from the source gNB, wherein the frequency list including a list of one or more Uu direct links and one or more PC5 sidelinks; a response handler that sends a measurement report upon detecting a predefined path switching triggering event based on the received measurement configuration; a command receiver that receives a path switching message from the source gNB indicating to switch from the sidelink relay path to a target direct Uu link; and a path switch handler that performs a path switching with service continuity by establishing a connection with the target direct Uu link based on the path switching message.
17. The UE of claim 16, wherein the predefined path switching triggering event is a measurement of the target direct Uu link is an offset better than a measurement of the serving sidelink with the serving relay UE.
18. The UE of claim 16, wherein the path switching message is a RRC Reconfiguration message from the source gNB.
19. The UE of claim 18, wherein the RRC Reconfiguration message includes one or more elements comprising: a list of data radio bearers (DRBs) to be released, a RadioBearerConfig information element (IE) with an index of relay UE-remote UE association, a new IE with an index of relay UE-remote UE association to indicate one or more relay radio bearers to be released.
20. The UE of claim 16, wherein the direct target Uu link is with a new gNB, and wherein the path switching message is a RRC Reconfiguration message from the source gNB embedded a Handover Command.